

GOSFORD HIGH SCHOOL



2009

Trial HSC

MATHEMATICS EXTENSION 2

Time Allowed: 3 Hours + 5 minutes reading time

General Instructions:

- Reading Time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.
- Each question should be started in a separate writing booklet.

TOTAL MARKS – 120

- Attempt Questions 1 – 8
- All questions are of equal value.

QUESTION 1: (Use a separate Writing Booklet)

(a) Evaluate $\int_0^4 \frac{x}{\sqrt{2x+1}} dx$ (3)

(b)

(i) Use the substitution $x = \frac{\pi}{2} - u$ to show that:

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x}{\sin x + \cos x} dx \quad (2)$$

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$. (2)

(c)

(i) Express $\frac{8}{(x+2)(x^2+4)}$ in the form $\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$ (2)

(ii) Show that $\int_0^2 \frac{8}{(x+2)(x^2+4)} dx = \frac{1}{2} \log_e 2 + \frac{\pi}{4}$ (2)

(d) Given that $I_n = \int_0^1 x^n e^x dx$ for $n=0,1,2,\dots$

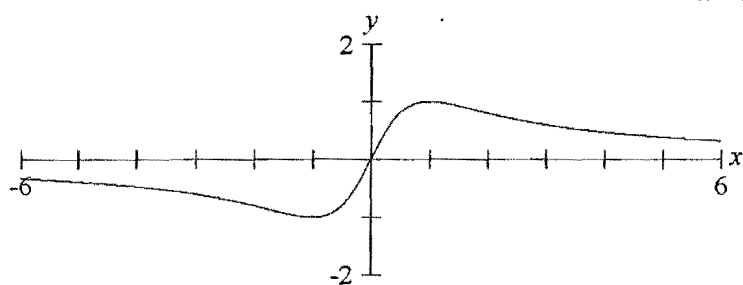
(i) Find I_0 . (1)

(ii) Find an expression for I_n in terms of I_{n-1} for $n=0,1,2,\dots$ (2)

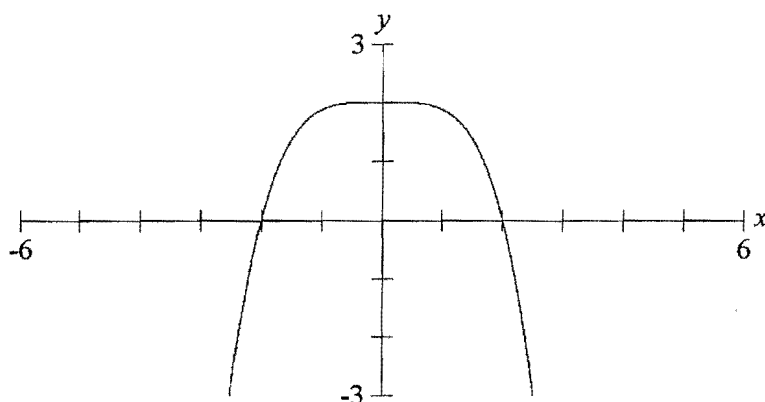
(iii) Evaluate I_3 . (1)

QUESTION 2: (Use a separate Writing Booklet)

(a) The diagrams below represent the curves $f(x) = \frac{2x}{x^2+1}$ and $g(x) = 2 - \frac{x^4}{8}$.



$$f(x) = \frac{2x}{x^2+1}$$



$$g(x) = 2 - \frac{x^4}{8}$$

Use these diagrams to sketch the following functions (without calculus) showing all essential features.

(i) $y = f(-x)$ (1)

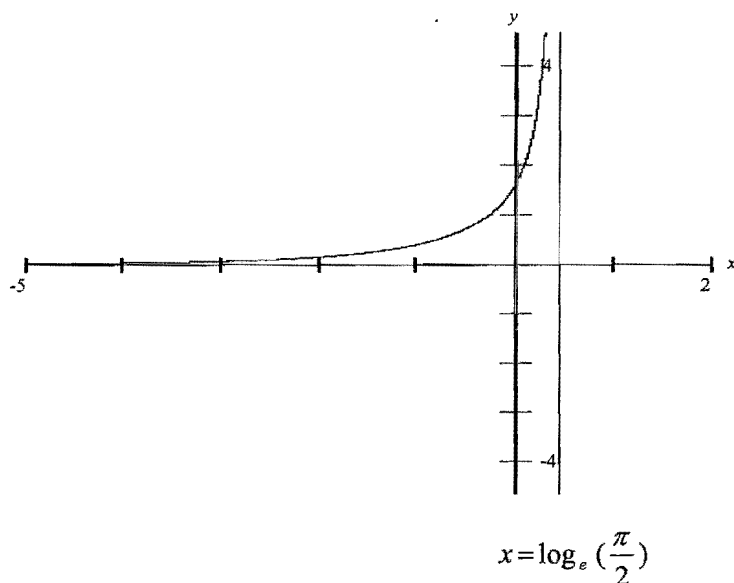
(ii) $y = |f(x)|$ (1)

(iii) $y = \sqrt{f(x)}$ (1)

(iv) $y = \frac{1}{g(x)}$ (2)

(v) $y = [g(x)]^2$ (2)

- (b) The diagram below shows part of the curve $y = \tan(e^x)$ where $x < \log_e\left(\frac{\pi}{2}\right)$.
The part to the right has not yet been drawn.



- (i) By considering values of x greater than $x = \log_e\left(\frac{\pi}{2}\right)$ find the smallest possible solution to the equation $\tan(e^x) = 0$. (1)
- (ii) Copy the diagram and then sketch the curve $y = \tan(e^x)$ for $\log_e\left(\frac{\pi}{2}\right) < x < \log_e\left(\frac{3\pi}{2}\right)$. (2)
- (iii) How many solutions are there to $\tan(e^x) = 0$ in the domain $1 < x < 3$? (2)
- (iv) Find the equation of the inverse function of $y = \tan(e^x)$ for the case where $x < \log_e\left(\frac{\pi}{2}\right)$ and draw a neat sketch of this curve. (3)

QUESTION 3: (Use a separate Writing Booklet)

(a) The complex number z is given by $z = 1 + \frac{1+i}{1-i}$.

(i) Express z in the form $a+ib$, where a & b are real. (1)

(ii) Find

(α) $\operatorname{Re}(z^2)$. (1)

(β) $|z|$ and $\arg(z)$. (1)

(γ) z^5 in the form $x+iy$, where x & y are real. (2)

(b)

(i) On an Argand diagram sketch the locus of $|z|=1$ & $|z-1|=1$ (1)

(ii) Hence, or otherwise, find in the form $a+ib$, where a & b are real, all complex numbers simultaneously satisfying $|z|=1$ & $|z-1|=1$. (2)

(c)

(i) Solve $z^3 - 1 = 0$ giving your answers in modulus-argument form. (1)

(ii) Let ω be one of the non-real roots of $z^3 - 1 = 0$.

(α) Show that $1 + \omega + \omega^2 = 0$ (1)

(β) Hence simplify $(1 + \omega)^5$ (1)

(d)

(i) Find the Cartesian equation and sketch the locus of z if $|z-i| = \operatorname{Im}(z)$ (2)

(ii) What is the least value of $\arg(z)$ in part (i)? (2)

QUESTION 4: (Use a separate Writing Booklet)

(a) If $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ has a triple root, find all the roots. (4)

(b) α, β, γ are the roots of $2x^3 - 4x^2 - 3x - 1 = 0$.

(i) Show that $(\alpha - 1)(\beta - 1)(\gamma - 1) = 3$ (2)

(ii) Hence find the value of $(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)(\alpha + \beta - \gamma)$ (2)

(c)

(i) Given that $z = \cos \theta + i \sin \theta$, use De Moivre's theorem to show that:

$$z^n + z^{-n} = 2 \cos n\theta \quad (2)$$

(ii) Hence, or otherwise, solve the equation

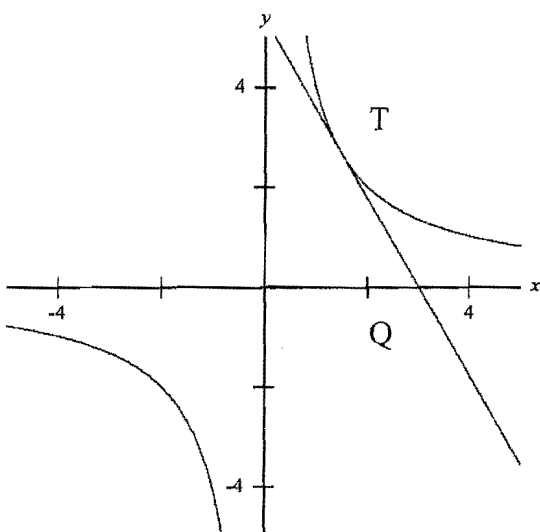
$$2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0 \quad (5)$$

QUESTION 5: (Use a separate Writing Booklet)

(a) The hyperbola H has equation $9x^2 - 16y^2 = 144$. Find the eccentricity, the coordinates of its foci, the equation of each directrix and the equation of each asymptote. Sketch the curve and indicate the foci, directrices and asymptotes. (5)

(b) An ellipse E has equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Given that $x = 4 \cos \theta$ & $y = 3 \sin \theta$ are parametric equations of E, derive the equations of the tangent and normal to the ellipse when $\theta = \frac{\pi}{3}$. (5)

(c) The tangent to the rectangular hyperbola $xy = 4$ at the point T $(2t, \frac{2}{t})$ has equation $x + t^2y = 4t$. The tangent cuts the x-axis at Q.

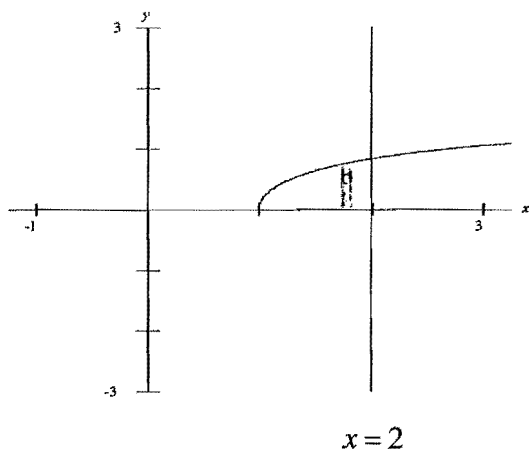


(i) Show that the line through Q, which is perpendicular to the tangent at T, has equation $t^2x - y = 4t^3$. (2)

(ii) This line cuts the rectangular hyperbola at the points R and S. Find the locus of M, the midpoint of RS, in Cartesian form. (3)

QUESTION 6: (Use a separate Writing Booklet)

- (a) By taking slices perpendicular to the x-axis find the volume obtained by rotating the region bounded by the curve $y = \sqrt{\log_e x}$ and the line $x=2$ about the x-axis. (4)



(b)

- (i) The region bounded by the curve $y = (x-1)^2$ and the x and y-axes is rotated through 360° about the line $y = -\frac{1}{2}$ to form a solid. If a vertical line segment is drawn from the point $P(x, y)$ on the curve, where $0 < x < 1$, to the x-axis it sweeps out an annulus. Show that the area of the annulus is given by:

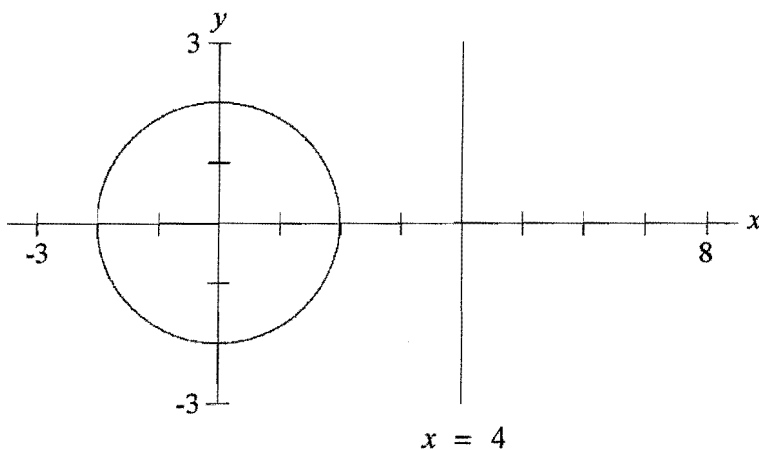
$$A = \pi [(x-1)^4 + (x-1)^2]. \quad (3)$$

- (ii) Hence find the volume of the solid. (2)

(c)

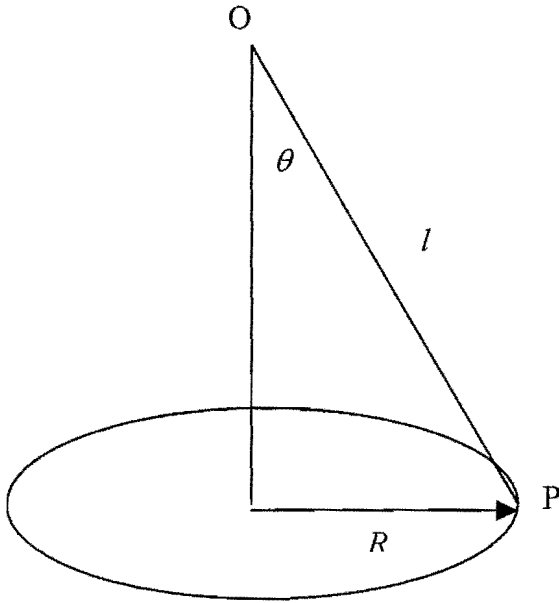
- (i) Use the substitution $x = 2 \sin \theta$ to evaluate $\int_{-2}^2 \sqrt{4-x^2} dx$ (3)

- (ii) Find the volume of the solid formed by rotating the circle $x^2 + y^2 = 4$ about the line $x=4$ using the method of cylindrical shells. (4)



QUESTION 7: (Use a separate Writing Booklet)

(a) A particle P of mass m kg is suspended from the end of a light inelastic string of length l metres which is fixed at a point O. The particle is moving with constant angular velocity ω and describes a circle of radius R metres in the horizontal plane.

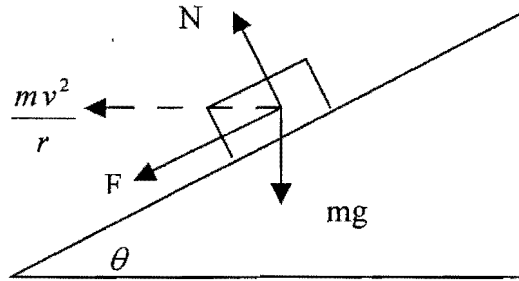


(i) By considering the forces acting on the particle P show that the angle θ between the string and the vertical through O is given by $\tan \theta = \frac{R\omega^2}{g}$. (2)

(ii) Show that the period of rotation is given by $2\pi \sqrt{\frac{l \cos \theta}{g}}$ (2)

(iii) A particle of mass 600 grams is attached to a light inelastic string, fixed at O, and moves uniformly in a horizontal circle with a period of 1.7 seconds. If the tension in the string is 20 newtons find the length of the string correct to 2 decimal places given that $g = 9.8 \text{ m s}^{-2}$. (3)

(b) A train line is banked at an angle of θ to the horizontal as shown below.



A train of mass m is travelling at a constant speed v in a horizontal circular arc of radius r on the banked train line.

- (i) If the force due to gravity is mg and the force of circular motion is given as $\frac{mv^2}{r}$ find the frictional force F and the normal reaction force N in terms of m , v , r , g and θ . (4)

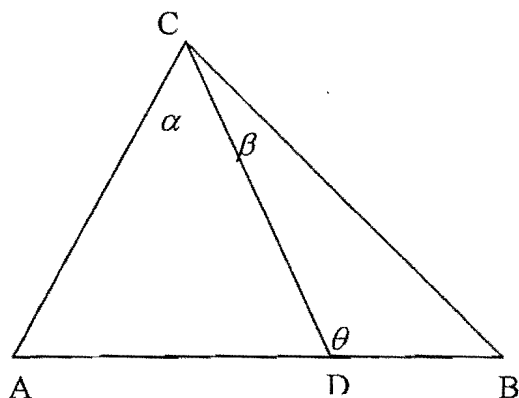
- (ii) A train rounding a banked circular bend of radius 500 metres exerts the same frictional force along the slope when travelling at 30 km/h as it does when travelling at 90 km/h but in opposite directions. Assume that acceleration due to gravity g is approximately 10 m s^{-2} .

- (α) Find the angle θ at which the train line is banked correct to the nearest minute. (3)

- (β) Find the optimum speed that the train should travel at so that $F = 0$. (Give your answer correct to the nearest km/h.) (1)

QUESTION 8: (Use a separate Writing Booklet)

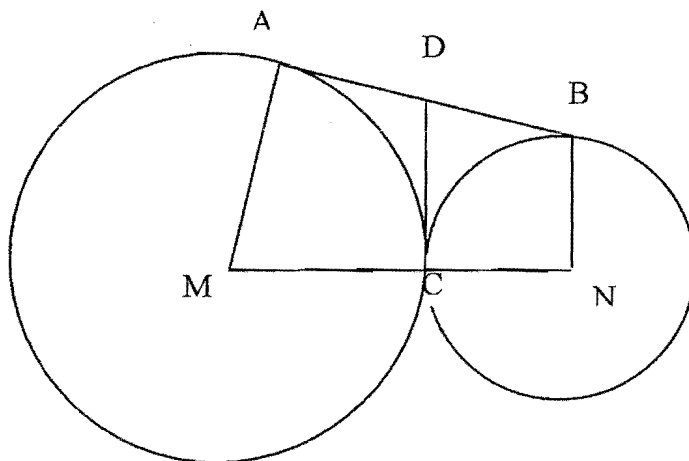
(a)



In $\triangle ABC$, D is the point on AB that divides AB internally in the ratio $m : n$. If $\angle ACD = \alpha$, $\angle BCD = \beta$ and $\angle CDB = \theta$, by using the sine rule in each of the triangles CAD and CDB , show that

$$\frac{\sin(\theta + \beta) \sin \alpha}{\sin(\theta - \alpha) \sin \beta} = \frac{m}{n} \quad (4)$$

(b)



In the diagram MCN is a straight line. Circles are drawn with centres M and N and radii MC and NC respectively. AB is a common tangent to the two circles with points of contact at A and B respectively. CD is a common tangent at C and meets AB at D .

(i) Show that $AMCD$ and $BNCD$ are cyclic quadrilaterals. (2)

(ii) Prove that $\triangle ACD$ is similar to $\triangle CBN$. (3)

(iii) Prove that $MD \parallel CB$. (1)

(c)

(i) By finding the equation of the tangent to $y = \log_e x$ at the point where $x=1$ prove that for $x > 0$, $\log_e x \leq x-1$ (3)

(ii) If $\{p_1, p_2, \dots, p_n\}$ is a set of n positive numbers adding to unity, i.e.

$p_1 + p_2 + \dots + p_n = 1$ and each $p_r > 0$, prove that $\sum_{r=1}^n \log_e (n p_r) \leq 0$. (2)

END OF PAPER.

Q1

a) $\int_0^4 \frac{x}{\sqrt{2x+1}} dx$

Let $u = \sqrt{2x+1}$
 $= (2x+1)^{1/2}$
 $du = \frac{1}{2} (2x+1)^{-1/2} \cdot 2 dx$

$u^2 = 2x+1$
 $u^2 - 1 = 2x$
 $x = \frac{u^2 - 1}{2}$

ie $du = \frac{1}{\sqrt{2x+1}} dx$

When $x=0$, $u=1$
 When $x=4$, $u=3$

$\therefore I = \int_1^3 \frac{u^2 - 1}{2} du$

$= \frac{1}{2} \left[\frac{u^3}{3} - u \right]_1^3$

$= \frac{1}{2} \left[\left(\frac{27}{3} - 3 \right) - \left(\frac{1}{3} - 1 \right) \right]$

$= \frac{1}{2} \times \frac{2}{3}$

$= \frac{1}{3}$

b) (i) Let $I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$

Let $x = \frac{\pi}{2} - u$

When $x=0$, $u = \frac{\pi}{2}$

$dx = -du$

When $x = \frac{\pi}{2}$, $u=0$

$\therefore I = \int_{\pi/2}^0 \frac{\sin(\frac{\pi}{2} - u)}{\sin(\frac{\pi}{2} - u) + \cos(\frac{\pi}{2} - u)} (-du)$

$= - \int_0^{\pi/2} \frac{\cos u}{\cos u + \sin u} du$

$= \int_0^{\pi/2} \frac{\cos u}{\cos u + \sin u} du$

$= \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$

(ii) $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx + \int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} dx$

$= \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx$

$= \int_0^{\pi/2} 1 dx$

$= \left[x \right]_0^{\pi/2}$

$= \frac{\pi}{2}$

$\therefore$ Since $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} dx$

$2I = \frac{\pi}{2}$

$I = \frac{\pi}{4}$

c) (i) $I \left\{ \frac{8}{(x+2)(x^2+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+4} \right.$

$8 = A(x^2+4) + (Bx+C)(x+2)$

Let $x = -2$,

$8 = A(8) + 0$

$A = 1$

Let $x=0$,

$$8 = 4A + 2C$$

$$8 = 4 + 2C$$

$$\therefore C = 2$$

Let $x=1$,

$$8 = 5A + (B+2)3$$

$$8 = 5 + 3B + 6$$

$$\therefore B = -1$$

$$\therefore \frac{8}{(x+2)(x^2+4)} = \frac{1}{x+2} + \frac{-x+2}{x^2+4}$$

$$\text{ii) } \int_0^2 \frac{8}{(x+2)(x^2+4)} dx = \int_0^2 \left(\frac{1}{x+2} + \frac{-x}{x^2+4} + \frac{2}{x^2+4} \right) dx$$

$$= \left[\ln(x+2) - \frac{1}{2} \ln(x^2+4) + \tan^{-1}\left(\frac{x}{2}\right) \right]_0^2$$

$$= \left[\ln 4 - \frac{1}{2} \ln 8 + \frac{\pi}{4} \right] - \left[\ln 2 - \frac{1}{2} \ln 4 + 0 \right]$$

$$= 2\ln 2 - \frac{3}{2} \ln 2 + \frac{\pi}{4} - \ln 2 + \ln 2 - 0$$

$$= \frac{1}{2} \ln 2 + \frac{\pi}{4}$$

d) (i) $I_0 = \int_0^1 e^x dx$

$$= [e^x]_0^1$$

$$= e - 1$$

(iii) $I_n = \int_0^1 x^n e^x dx$

$$u = x^n \quad v' = e^x$$

$$u' = nx^{n-1}$$

$$v = e^x$$

$$\therefore I_n = [x^n e^x]_0^1 - n \int_0^1 x^{n-1} e^x dx$$

$$I_n = 1 \cdot e^1 - 0 \cdot e^0 - n I_{n-1}$$

$$= e - n I_{n-1}$$

(iii)

$$I_3 = e - 3 I_2$$

$$= e - 3 [e - 2 I_1]$$

$$= e - 3 [e - 2(e - 1 I_0)]$$

$$= e - 3 [e - 2(e - (e - 1))]$$

$$= e - 3 [e - 2(e - e + 1)]$$

$$= e - 3 [e - 2]$$

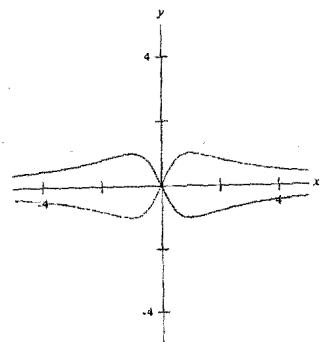
$$= e - 3e + 6$$

$$= 6 - 2e$$

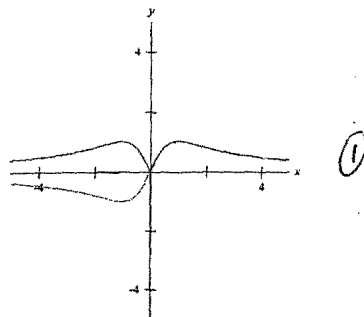
Q2

a)

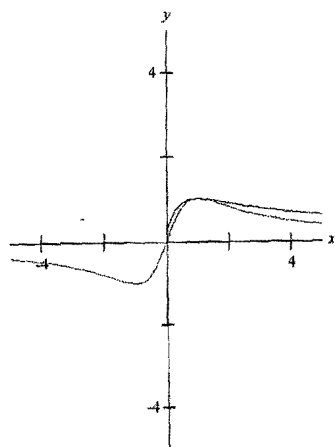
(i)



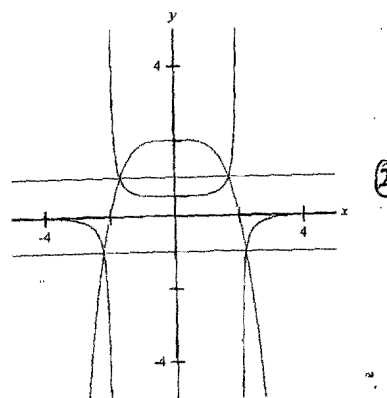
(ii)



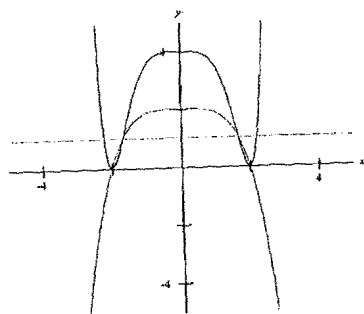
(iii)



(iv)



(v)



b)

$$(i) \quad \int \tan(e^x) = 0$$

$$e^x = 0, \pi, 2\pi, \dots$$

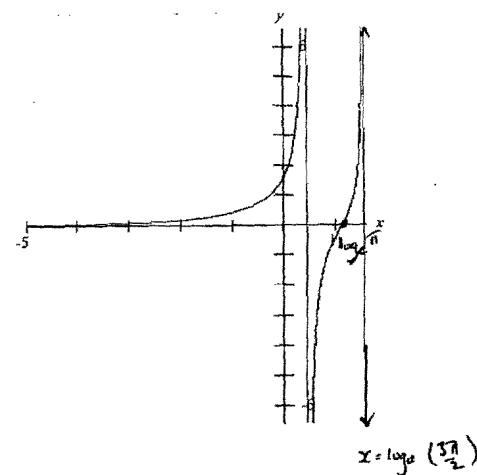
$\therefore$ the smallest positive solution is

$$e^x = \pi$$

$$\text{i.e. } x = \log_e \pi$$

①

(ii)



②

$$(iii) \quad \int \tan(e^x) = 0$$

$$e^x = 0, \pi, 2\pi, 3\pi, \dots$$

$$\text{Now } \ln \pi \approx 1.14, \quad \ln 2\pi \approx 2.94, \quad \ln 3\pi \approx 3.09$$

$$\therefore x = \log_e \pi, \log_e 2\pi, \dots, \log_e 6\pi, \text{ for } 1 < x$$

Hence there are 6 solutions.

②

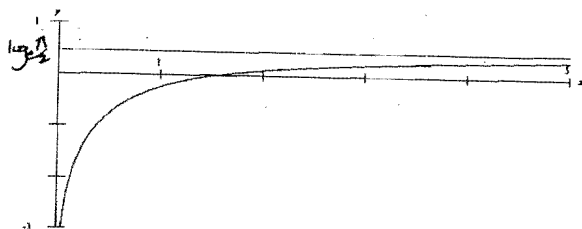
$$(iv) \quad y = \tan(e^x) \text{ is a one to one function for } x < \log_e \left(\frac{\pi}{2}\right)$$

$\therefore$ Its inverse is given by

$$x = \tan(e^y), \quad y < \log_e(\frac{\pi}{2})$$

$$\therefore e^y = \tan^{-1}(x)$$

$$y = \log_e(\tan^{-1}(x)) \quad (2)$$



(1)

Q3

$$(i) z = 1 + \frac{1+i}{1-i} \times \frac{1+i}{1+i}$$

$$= 1 + \frac{(1+i)^2}{1-i^2}$$

$$= 1 + \frac{1+2i+i^2}{2}$$

$$= \frac{2 + 1+2i-1}{2}$$

$$= 1+i$$

(1)

$$(ii) (d) \quad z^2 = (1+i)^2$$

$$= 1+2i+i^2$$

$$= 2i$$

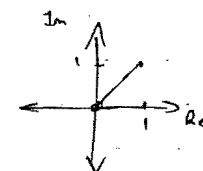
$$\therefore \operatorname{Re}(z^2) = 0$$

(1)

$$(b) \quad |z| = \sqrt{1+1}$$

$$= \sqrt{2}$$

$$\operatorname{Arg}(z) = \frac{\pi}{4}$$



(1)

$$(8) \quad \text{Now } z = \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right)$$

$$\therefore z^5 = (\sqrt{2})^5 \operatorname{cis}\left(\frac{5\pi}{4}\right)$$

$$= 4\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)$$

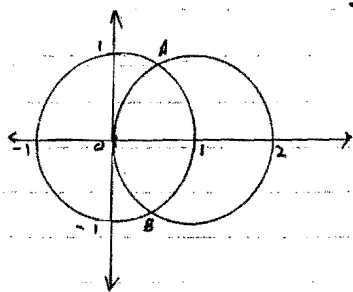
$$= 4\sqrt{2} \left(-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \right)$$

$$= 4(-1-i)$$

$$= -4 - 4i$$

(2)

b) (i)



(1)

(ii) Let the points of intersection be A & B
 & let the centres of the circles be O & C

ΔAOC is equilateral since $OA = OC = AC = 1$ unit

& ΔBOC " " " $OB = OC = BC = 1$ unit

$\therefore$ A is the pt $\text{cis}(\frac{\pi}{3})$ & B is the pt $\text{cis}(-\frac{\pi}{3})$

$\therefore$ The soln are $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ & $\frac{1}{2} - \frac{\sqrt{3}}{2}i$.

(2)

c) (i) Let $z = \text{cis } \theta$

$$\text{If } z^3 = 1$$

$$\text{cis } 3\theta = 1$$

$$3\theta = 0, 2\pi, 4\pi,$$

$$\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\therefore z = 1, \text{cis}(\frac{2\pi}{3}), \text{cis}(\frac{4\pi}{3})$$

(1)

(ii) Let the roots be $1, \omega, \omega^2$

$$\text{For } z^3 - 1 = 0$$

$$\sum \alpha = 1 + \omega + \omega^2 = -\frac{b}{a}$$

$$= 0$$

(1)

$$\begin{aligned} b) (1+\omega)^5 &= (-\omega^2)^5 \\ &= -(\omega^2)^{10} \\ &= (\omega^3)^3 \times -\omega \end{aligned}$$

Since ω is a solution to $z^3 - 1 = 0$
 $\omega^3 = 1$.

(1)

$$\begin{aligned} \therefore (1+\omega)^5 &= 1^3 \times -\omega \\ &= -\omega \end{aligned}$$

d) Let $z = x + iy$

$$|z - i| = |x + i(y - 1)|$$

$$= \sqrt{x^2 + (y-1)^2}$$

$$\text{Im}(z) = y$$

$$\therefore y = \sqrt{x^2 + (y-1)^2}$$

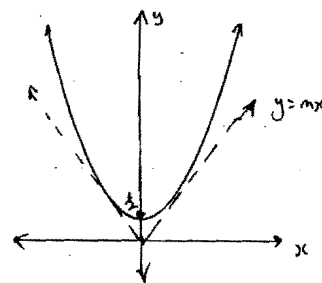
$$y^2 = x^2 + (y-1)^2$$

$$y^2 = x^2 + y^2 - 2y + 1$$

$$\therefore 2y = x^2 + 1$$

$$y = \frac{x^2 + 1}{2}$$

(2)



(ii) Let $y = mx$ be a tangent to $y = \frac{x^2+1}{2}$

pt of intersection is given by

$$\frac{x^2+1}{2} = mx$$

$$x^2 - 2mx + 1 = 0$$

If $y = mx$ is a tangent $\Delta = 0$

$$\text{i.e. } 4m^2 - 4 = 0$$

$$m^2 = 1$$

$$m = \pm 1$$

(2)

$\therefore$ the least value of $\arg(z)$ is $\tan^{-1}(1)$

$$\text{i.e. } \frac{\pi}{4}$$

Q4

a) Let $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

$$\text{If } P'(x) = 0$$

$$24x^2 + 54x + 12 = 0$$

$$4x^2 + 9x + 2 = 0$$

$$(4x+1)(x+2) = 0$$

$$x = -\frac{1}{4} \text{ or } -2$$

$$\text{Now } P'(-\frac{1}{4}) = 8(-\frac{1}{4})^3 + 27(-\frac{1}{4})^2 + 12(-\frac{1}{4}) - 20 \neq 0$$

$$\text{But } P'(-2) = 8(-2)^3 + 27(-2)^2 + 12(-2) - 20 = 0$$

$\therefore -2$ is the triple root.

(4)

the sum of the roots $\sum \alpha = -\frac{b}{a}$

$$\therefore -2 + -2 + -2 + k = -\frac{9}{2}$$

$$-6 + k = -\frac{9}{2}$$

$$k = \frac{3}{2}$$

$\therefore$ the roots are $-2, -2, -2, \frac{3}{2}$

$$\text{b) (i) } (\alpha-1)(\beta-1)(\gamma-1) = (\alpha-1)[\beta\gamma - \beta - \gamma + 1]$$

$$= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$$

$$= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1$$

$$= -\frac{1}{2} - \frac{1}{2} + \frac{5}{2} - 1$$

$$= \frac{1}{2} - \frac{3}{2} + 2 - 1$$

$$= 3$$

(2)

(ii) Since $\alpha + \beta + \gamma = 2$

$$\alpha + \beta - \gamma = 2 - 2\gamma$$

Similarly $\beta + \gamma - \alpha = 2 - 2\alpha$

$$\alpha + \gamma - \beta = 2 - 2\beta$$

$$\begin{aligned} & \therefore (\beta + \gamma - \alpha)(\gamma + \alpha - \beta)(\alpha + \beta - \gamma) \\ &= (2 - 2\alpha)(2 - 2\beta)(2 - 2\gamma) \\ &= -2(\alpha - 1) \cdot -2(\beta - 1) \cdot -2(\gamma - 1) \\ &= -8(\alpha - 1)(\beta - 1)(\gamma - 1) \\ &= -8 \times 3 \\ &= -24 \end{aligned}$$

(2)

(*) c) i) If $z = \cos \theta + i \sin \theta$
 $z^n = \cos(n\theta) + i \sin(n\theta)$

$$\begin{aligned} \& z^{-n} &= \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta - i \sin(n\theta) \quad \text{since } \cos(-A) = \cos A \\ &\& \sin(-A) = -\sin A \end{aligned}$$

$$\therefore z^n + z^{-n} = 2 \cos(n\theta)$$

ii) If $2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$

$$z^2(2z^2 + 3z + 5 + 3z^{-1} + 2z^{-2}) = 0$$

$$z^2[2z^2 + 2z^{-2} + 3z + 3z^{-1} + 5] = 0$$

$$z^2[2(2\cos 2\theta) + 3(2\cos \theta) + 5] = 0$$

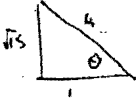
$$z^2[4(2\cos^2 \theta - 1) + 6\cos \theta + 5] = 0$$

$$z^2[8\cos^2 \theta + 6\cos \theta + 1] = 0$$

$$s_0 \quad z^2[(4\cos \theta + 1)(2\cos \theta + 1)] = 0$$

$$\therefore z^2 = 0, \quad \cos \theta = -\frac{1}{4}, \quad \cos \theta = -\frac{1}{2}$$

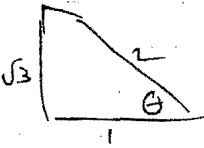
If $\cos \theta = -\frac{1}{4}$



$\sqrt{S}$	A
$\sqrt{T}$	C

$$\sin \theta = \pm \frac{\sqrt{15}}{4}$$

If $\cos \theta = -\frac{1}{2}$



$\sqrt{S}$	A
$\sqrt{T}$	C

$$\sin \theta = \pm \frac{\sqrt{3}}{2}$$

If $z^2 = 0$, $\cos 2\theta + i \sin 2\theta = 0 + 0i$
 $\therefore \cos 2\theta = 0$ & $\sin 2\theta = 0$
 No soln.

$$\therefore z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i, \quad -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

(5)

Q5

a) $9x^2 - 16y^2 = 144$

$$\frac{9x^2}{144} - \frac{16y^2}{144} = 1$$

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

$\therefore a=4, b=3$

Now $b^2 = a^2(e^2 - 1)$

$$\frac{9}{16} = e^2 - 1$$

$$e^2 = \frac{25}{16}$$

$$e = 5/4$$

The eccentricity is $5/4$.

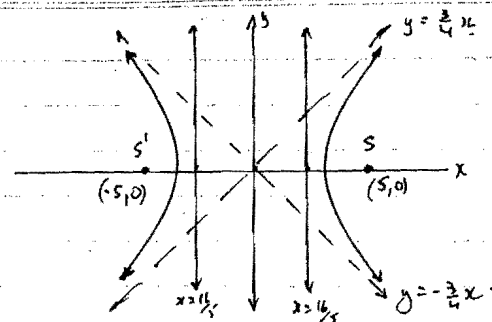
$$ae = 4 \times \frac{5}{4} = 5$$

The foci are $(5,0)$ & $(-5,0)$

$$\frac{a}{e} = \frac{4}{5/4} = \frac{16}{5}$$

The eqⁿ of the directrices are $x = \pm \frac{16}{5}$

The eqⁿ of the asymptotes are $y = \pm \frac{3}{4}x$



b) If $x = 4 \cos \theta$ $\frac{dx}{d\theta} = -4 \sin \theta$

If $y = 3 \sin \theta$ $\frac{dy}{d\theta} = 3 \cos \theta$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{d\theta} \times \frac{d\theta}{dx} \\ &= 3 \cos \theta \times \frac{-1}{4 \sin \theta} \\ &= -\frac{3}{4} \cot \theta \end{aligned}$$

When $\theta = \frac{\pi}{3}$,

$$\begin{aligned} \frac{dy}{dx} &= -\frac{3}{4} \cot \frac{\pi}{3}, \quad x = 4 \cos \frac{\pi}{3}, \quad y = 3 \sin \frac{\pi}{3} \\ &= -\frac{3}{4} \cdot \frac{1}{\sqrt{3}} \quad = 4 \cdot \frac{1}{2} \quad = 3 \cdot \frac{\sqrt{3}}{2} \\ &= -\frac{\sqrt{3}}{4} \quad = 2 \quad = \frac{3\sqrt{3}}{2} \end{aligned}$$

$\therefore$ the eqⁿ of the tangent is $y - \frac{3\sqrt{3}}{2} = -\frac{\sqrt{3}}{4}(x - 2)$

$$\begin{aligned} 4y - 6\sqrt{3} &= -\sqrt{3}x + 2\sqrt{3} \\ \text{i.e. } \sqrt{3}x + 4y - 8\sqrt{3} &= 0 \end{aligned}$$

The eqⁿ of the normal is of the form:

$$4x - \sqrt{3}y = k$$

$$\text{So } 4(2) - \sqrt{3}\left(\frac{3\sqrt{3}}{2}\right) = k$$

$$8 - \frac{9}{2} = k$$

$$k = \frac{7}{2}$$

$$\text{i.e. } 4x - \sqrt{3}y = \frac{7}{2}$$

$$\text{or } 8x - 2\sqrt{3}y - 11 = 0.$$

$$\text{d) i) If } x + t^2y = 4t$$

$$\text{When } y=0, \quad x=4t$$

$$\therefore O \text{ is the pt } (4t, 0)$$

$$\text{Also } x + t^2y = 4t$$

$$t^2y = -x + 4t$$

$$y = -\frac{1}{t^2}x + \frac{4}{t}$$

$$\therefore \text{the grad. of the } \perp \text{ to } TD \text{ is } t^2$$

the eqn of the normal is

$$y - 0 = t^2(x - 4t)$$

$$y = t^2x - 4t^3$$

$$\text{i.e. } t^2x - y = 4t^3$$

(ii) Pts of intersection occur when

$$t^2x - 4t^3 = \frac{4}{x}$$

$$\text{i.e. } t^2x^2 - 4t^3x - 4 = 0$$

If M is the point (X, Y)

$$X = \frac{x_1 + x_2}{2}$$

where x_1 & x_2 are the roots of (*)

$$\text{But } x_1 + x_2 = -\frac{b}{a}$$

$$= \frac{4t^3}{t^2}$$

$$= 4t$$

$$\therefore X = 2t$$

$$\text{If } X = 2t, \quad y = t^2x - 4t^3$$

$$= t^2 \times 2t - 4t^3$$

$$= 2t^3 - 4t^3$$

$$y = -2t^3$$

$$M \text{ is the pt } (2t, -2t^3)$$

$$\text{If } x = 2t$$

$$t = \frac{x}{2}$$

$$y = -2t^3$$

$$= -2\left(\frac{x}{2}\right)^3$$

$$= -\frac{x^3}{4}$$

$$\text{The locus of M is } y = -\frac{x^3}{4}$$

Q6

a) Let the width of each slice be Δx

$$\Delta V = \pi \cdot y^2 \cdot \Delta x$$

$$\therefore V = \lim_{\Delta x \rightarrow 0} \sum_1^2 \pi \cdot y^2 \cdot \Delta x$$

$$= \int_1^2 \pi y^2 dx$$

$$= \pi \int_1^2 \log_e x \, dx$$

$$u = \log_e x \quad v' = 1$$

$$u' = \frac{1}{x} \quad v = x$$

$$\therefore I = \pi \left\{ \left[x \log_e x \right]_1^2 - \int_1^2 \frac{1}{x} \cdot x \, dx \right\}$$

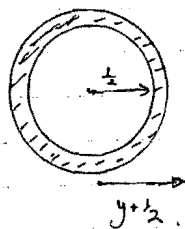
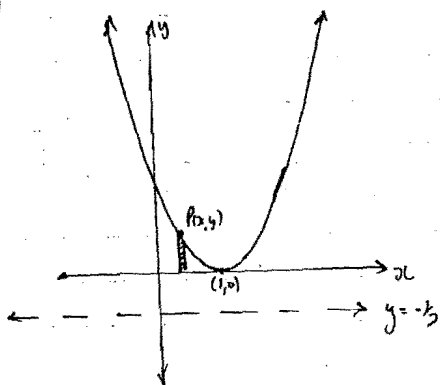
$$= \pi \left\{ (2 \log_e 2 - 2 \log_e 1) - [x]_1^2 \right\}$$

$$= \pi (2 \log_e 2 - (2-1))$$

$$= \pi (2 \log_e 2 - 1) \text{ units}^2$$

b)

(i)



$$A = \pi (R^2 - r^2) \quad \text{where } R = y + \frac{1}{2}, \quad r = \frac{1}{2}$$

$$\therefore A = \pi \left[\left(y + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \right]$$

$$= \pi \left[\left(y + \frac{1}{2} + \frac{1}{2}\right) \left(y + \frac{1}{2} - \frac{1}{2}\right) \right]$$

$$= \pi [(y+1) \cdot y]$$

$$= \pi [\{(x-1)^2 + 1\} \{x-1\}]$$

$$= \pi [(x-1)^4 + (x-1)^2]$$

$$(ii) \Delta V = \pi [(x-1)^4 + (x-1)^2] \cdot \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_0^1 \pi [(x-1)^4 + (x-1)^2] \cdot \Delta x$$

$$= \pi \int_0^1 (x-1)^4 + (x-1)^2 \, dx$$

$$= \pi \left[\frac{(x-1)^5}{5} + \frac{(x-1)^3}{3} \right]_0^1$$

$$= \pi \left[(0+0) - \left(-\frac{1}{5} + -\frac{1}{3}\right) \right]$$

$$= \pi \times \frac{8}{15}$$

$$= \frac{8\pi}{15} \text{ units}^3$$

c)

$$(i) \text{ If } x = 2 \sin \theta$$

$$dx = 2 \cos \theta \, d\theta$$

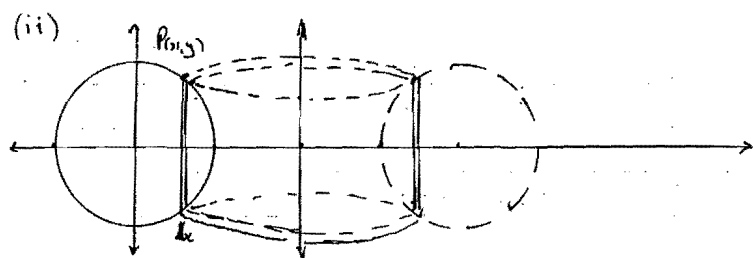
$$\text{When } x = -2, \quad \theta = -\frac{\pi}{2}$$

$$\text{When } x = 2, \quad \theta = \frac{\pi}{2}$$

$$\therefore I = \int_{-\pi/2}^{\pi/2} \sqrt{4 - 4 \sin^2 \theta} \cdot 2 \cos \theta \, d\theta$$

$$\begin{aligned}
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{4\cos^2\theta} \cdot 2\cos\theta \, d\theta \\
 &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\cos\theta \cdot 2\cos\theta \, d\theta \\
 &= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2\theta \, d\theta \\
 &= 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos 2\theta \, d\theta \\
 &= 4 \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\
 &= 4 \left\{ \left(\frac{\pi}{4} + \frac{1}{4} \sin \pi \right) - \left(-\frac{\pi}{4} + \frac{1}{4} \sin(-\pi) \right) \right\} \\
 &= 4 \left(\frac{\pi}{4} + 0 + \frac{\pi}{4} - 0 \right) \\
 &= 2\pi
 \end{aligned}$$

(3)



The radius of the inner shell is $4-x+\Delta x$
 " " " outer " " $4-x$
 " height " " shells " $2y$

$$\begin{aligned}
 \Delta V &= \pi (R^2 - r^2) \cdot h \\
 &= \pi (R+r)(R-r) \cdot h
 \end{aligned}$$

$$\begin{aligned}
 &= \pi \left[(4-x+\Delta x) + (4-x) \right] \left[(4-x+\Delta x) - (4-x) \right] \cdot 2y \\
 &= \pi \left[2(4-x) + \Delta x \right] \cdot \Delta x \cdot 2y \\
 \therefore V &= \lim_{\Delta x \rightarrow 0} \sum_{-2}^2 \pi \left[2(4-x) + \Delta x \right] \cdot \Delta x \cdot 2y \\
 &= 4\pi \int_{-2}^2 (4-x) \cdot \sqrt{4-x^2} \, dx \\
 &= 4\pi \int_{-2}^2 4\sqrt{4-x^2} - x\sqrt{4-x^2} \, dx \\
 &= 16\pi \int_{-2}^2 \sqrt{4-x^2} \, dx - 4\pi \int_{-2}^2 x\sqrt{4-x^2} \, dx
 \end{aligned}$$

Now if

$$\begin{aligned}
 f(x) &= x\sqrt{4-x^2} \\
 f(-x) &= -x\sqrt{4-x^2} \\
 &= -f(x)
 \end{aligned}$$

$\therefore f(x)$ is odd
 Hence $\int_{-2}^2 x\sqrt{4-x^2} \, dx = 0$

$$\begin{aligned}
 \therefore V &= 16\pi \int_{-2}^2 \sqrt{4-x^2} \, dx \\
 &= 16\pi \times 2\pi \\
 &= 32\pi \text{ units}^2
 \end{aligned}$$

(4)

$$\begin{aligned}\text{Period} &= \frac{2\pi}{\omega} \\ &= \frac{2\pi}{\sqrt{\frac{g}{L \cos \theta}}} \\ &= 2\pi \sqrt{\frac{L \cos \theta}{g}}\end{aligned}$$

(2)

(iii) Since $T \cos \theta = mg$

$$20 \cos \theta = 0.6 \times 9.8$$

$$\cos \theta = \frac{0.6 \times 9.8}{20}$$

$$\therefore 1.7 = 2\pi \sqrt{\frac{L \left(\frac{0.6 \times 9.8}{20} \right)}{9.8}}$$

$$1.7 = 2\pi \sqrt{L \left(\frac{0.6}{20} \right)}$$

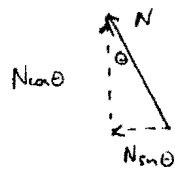
$$\left(\frac{1.7}{2\pi} \right)^2 = L \left(\frac{0.6}{20} \right)$$

$$L = \frac{\left(\frac{1.7}{2\pi} \right)^2}{0.03}$$

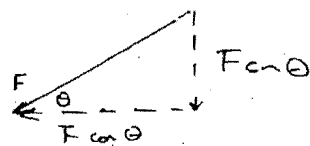
$$\doteq 2.46 \text{ metres (2.d.p.)}$$

b) (i) Forces acting on the train

N:

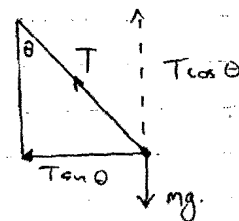


F:



07

(a) (i) Let T be the tension in the string



Resolving forces vertically and horizontally

$$V: T \cos \theta - mg = 0$$

$$H: T \sin \theta = mR\omega^2$$

$$\therefore T \cos \theta = mg \quad \text{and} \quad T \sin \theta = mR\omega^2$$

$$\frac{T \sin \theta}{T \cos \theta} = \frac{mR\omega^2}{mg}$$

(2)

$$\tan \theta = \frac{R\omega^2}{g}$$

$$(ii) \text{ If } \tan \theta = \frac{R\omega^2}{g}$$

$$\omega^2 = \frac{g \tan \theta}{R}$$

$$\text{But } \frac{R}{L} = \sin \theta$$

$$\begin{aligned}\therefore \omega^2 &= \frac{g \tan \theta}{L \sin \theta} \\ &= \frac{g \frac{\sin \theta}{\cos \theta}}{L \sin \theta} \\ &= \frac{g}{L \cos \theta}\end{aligned}$$

Resolving forces vertically & horizontally

$$H: N \sin \theta + F \cos \theta = \frac{mv^2}{r} \quad \text{--- (1)}$$

$$V: N \cos \theta - F \sin \theta = mg \quad \text{--- (2)}$$

$$(1) \times \cos \theta \quad \& \quad (2) \times \sin \theta$$

$$N \sin \theta \cos \theta + F \cos^2 \theta = \frac{mv^2}{r} \cos \theta \quad \text{--- (3)}$$

$$N \sin \theta \cos \theta - F \sin^2 \theta = mg \sin \theta \quad \text{--- (4)}$$

(3)-(4) gives

$$F \cos^2 \theta + F \sin^2 \theta = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

$$\therefore F = \frac{mv^2}{r} \cos \theta - mg \sin \theta \quad (2)$$

$$(1) \times \sin \theta \quad \& \quad (2) \times \cos \theta$$

$$N \sin^2 \theta + F \sin \theta \cos \theta = \frac{mv^2}{r} \sin \theta \quad \text{--- (5)}$$

$$N \cos^2 \theta - F \sin \theta \cos \theta = mg \cos \theta \quad \text{--- (6)}$$

(5)+(6) gives

$$N \sin^2 \theta + N \cos^2 \theta = \frac{mv^2}{r} \sin \theta + mg \cos \theta$$

$$\therefore N = \frac{mv^2}{r} \sin \theta + mg \cos \theta \quad (2)$$

(ii)

$$(a) \quad 30 \text{ km/hr} = 30 \times 1000 \div 60 \div 60 \text{ m s}^{-1}$$

$$= \frac{25}{3} \text{ m s}^{-1}$$

$$\therefore 90 \text{ km/hr} = \frac{75}{3} \text{ m s}^{-1}$$

Let the frictional force at 30 km/hr be F_1
 " " " " 90 km/hr " F_2

$$F_1 = \frac{m \left(\frac{25}{3} \right)^2 \cos \theta}{500} - m \cdot 10 \cdot \sin \theta$$

$$F_2 = \frac{m \left(\frac{75}{3} \right)^2 \cos \theta}{500} - m \cdot 10 \cdot \sin \theta$$

$$F_1 + F_2 = 0$$

$$\therefore m \cos \theta \left[\frac{\left(\frac{25}{3} \right)^2 + \left(\frac{75}{3} \right)^2}{500} \right] - 20m \sin \theta = 0$$

$$\therefore 20 \tan \theta = \frac{\left(\frac{25}{3} \right)^2 + \left(\frac{75}{3} \right)^2}{500} \quad (3)$$

$$\tan \theta = \frac{\left(\frac{25}{3} \right)^2 + \left(\frac{75}{3} \right)^2}{10000}$$

$$\theta = 3^\circ 58' \text{ nearest min.}$$

(B) If $F = 0$

$$\frac{mv^2}{r} \cos \theta = mg \sin \theta$$

$$\tan \theta = \frac{v^2}{rg} \quad (1)$$

$$v^2 = rg \tan \theta$$

$$v^2 = 500 \times 10 \times \tan 3^\circ 58'$$

$$v = 18.62 \text{ m s}^{-1}$$

$$= 67 \text{ km/hr (nearest km/hr)}$$

Q8

a) In $\triangle CAD$, $\angle CAD = \theta - \alpha$ (ext. $\angle$ of a $\triangle$ theorem)

$$\therefore \frac{CD}{\sin(\theta - \alpha)} = \frac{AD}{\sin \alpha}$$

$$AD = \frac{CD \sin \alpha}{\sin(\theta - \alpha)}$$

In $\triangle CDB$, $\angle CBD = 180 - (\theta + \beta)$ ($\angle$ sum of a $\triangle$ is 180°)

$$\therefore \frac{CD}{\sin[180 - (\theta + \beta)]} = \frac{DB}{\sin \beta}$$

$$DB = \frac{CD \sin \beta}{\sin[180 - (\theta + \beta)]}$$

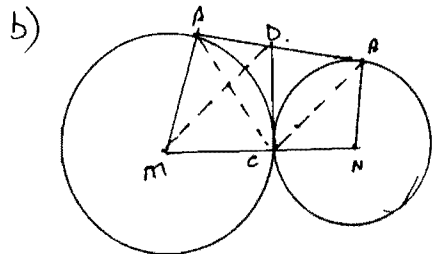
$$= \frac{CD \sin \beta}{\sin(\theta + \beta)} \quad \text{since } \sin(180 - A) = \sin A$$

$$\therefore \frac{AD}{DB} = \frac{\frac{CD \sin \alpha}{\sin(\theta - \alpha)}}{\frac{CD \sin \beta}{\sin(\theta + \beta)}}$$

$$\therefore \frac{m}{n} = \frac{\cancel{CD} \sin \alpha}{\sin(\theta - \alpha)} \times \frac{\sin(\theta + \beta)}{\cancel{CD} \sin \beta}$$

$$\therefore \frac{\sin(\theta + \beta) \sin \alpha}{\sin(\theta - \alpha) \sin \beta} = \frac{m}{n}$$

(4)



(i) $\angle DAM = \angle DCM = 90^\circ$ (tangent $\perp$ radius at pt of contact)

$\angle DBN = \angle DCN = 90^\circ$ (" " " " " ")

$\therefore \angle DAM = \angle DCM$ from a pair of suppl. opp. $\angle$'s

$\angle DBN = \angle DCN$ " " " " " "

Hence $AMCD$ & $BNCD$ are cyclic quads.

(2)

(ii) Join A to C & B to C

Let $\angle ADC$ be θ

$\therefore \angle BNC = \theta$ (ext. $\angle$ of a cyclic quad theorem)

Now $AD = DC$ (tangents from an external pt are equal in length)

$\therefore \triangle ADC$ is isosceles.

$\therefore \angle DAC = \angle DCA = 90 - \frac{\theta}{2}$ ($\angle$ sum of a $\triangle$ is 180° & base $\angle$'s of an isosc. $\triangle$ are equal)

Also $BN = CN$ (equal radii)

$\therefore \triangle BNC$ is isosceles.

(3)

$\therefore \angle NCB = \angle NBC$ ($\angle$ sum of a $\triangle$ is 180° & base $\angle$'s of an isosc. $\triangle$ are equal)

$\therefore \triangle ACD \parallel \triangle CBN$ ($\triangle$'s are equiangular)

(iii) Join M to D

Since $ADMC$ is a cyclic quad.

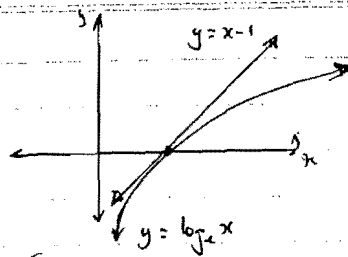
$\angle CMD = \angle CAD = 90 - \frac{\theta}{2}$ ($\angle$'s in the same segment are equal)

$\therefore \angle CMD = \angle NCB$ from a pair of equal corresp. $\angle$'s.

(1)

Hence $MD \parallel CB$

c) (i) If $y = \log_e x$
 $y' = \frac{1}{x}$



When $x=1$, $y'=1$, $y=0$

∴ The eqⁿ of the tangent at $(1, 0)$ is

$$y - 0 = 1(x - 1)$$

$$y = x - 1$$

Since $y = x - 1$ lies above $y = \log_e x$

$$\log_e x \leq x - 1 \text{ with equality when } x = 1$$

(3)

(ii) Since $p_r > 0$

$$\log_e (np_r) \leq np_r - 1$$

$$\log_e (np_2) \leq np_2 - 1$$

$$\log_e (np_n) \leq np_n - 1$$

Adding these we get

$$\sum_{r=1}^n \log_e (np_r) \leq np_1 - 1 + np_2 - 1 + \dots + np_n - 1$$

$$= n(p_1 + p_2 + \dots + p_n) - 1 \times n$$

$$= n \times 1 - n$$

$$= 0$$

$$\therefore \sum_{r=1}^n \log_e (np_r) \leq 0 \quad (2)$$